**Name:** _____

**Class:** _____

**Date:** _____ **Mr. Rawson • GCSE Physics**

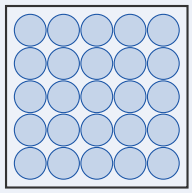
# 3.2 States of Matter

*In RP17 you measured density, $\rho = m/V$. Today: why the three states of matter have such different densities.*

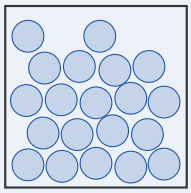
## 1 · The particle model — everything is made of tiny particles

**Key concept — the three states**

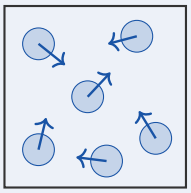
All matter is made of tiny **particles** (atoms or molecules). The three states differ in how the particles are **arranged**, how far **apart** they are, and how they **move**. The circles below are the particles; the box is the container.



**solid**



**liquid**



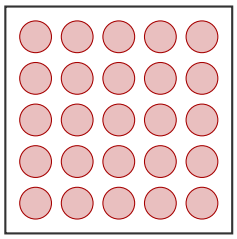
**gas**

**Solid**
- regular pattern, touching
- **vibrate** about fixed positions
- strong forces of attraction hold them in place

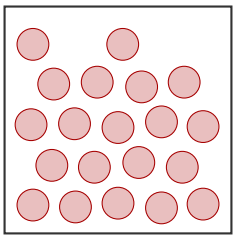
**Liquid**
- random, still touching
- **move around** each other
- forces strong enough to keep them together

**Gas**
- random, **far apart**
- move **fast** in all directions
- forces so weak the particles fly free

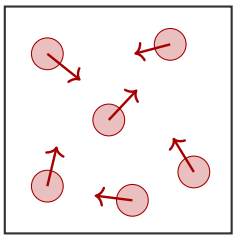
**1)** **Draw** the particles in a block of **ice**, in **liquid water**, and in **steam**. In the steam box, add an arrow to each particle to show its motion.



**ice (solid)**



**liquid water**



**steam (gas)**

**Check:** in your solid, every particle touches its neighbours and the rows line up. In your liquid, they still touch, and the rows are broken. In your gas, there is room for many more particles in the gaps than you have drawn, and every one has an arrow.

## 2 · Properties from the model — what the arrangement explains

**Each property of a state follows from its particles.** A solid keeps its shape because its particles are held in fixed positions. A liquid flows because its particles move around each other. A gas fills its container because its particles move freely in every direction, and it can be squashed because of the empty space between them.

**2) Complete** the table. Three cells are filled in as examples; write a short phrase in each of the others.

| | Solid | Liquid | Gas |
|---|---|---|---|
| **Arrangement** | regular pattern | *random, touching* | *random, far apart* |
| **Movement** | *vibrate about fixed positions* | move around each other | *fast, random, in all directions* |
| **Spacing** | *touching* | *touching* | far apart |
| **Shape** | *fixed shape* | *takes the shape of its container* | *fills its container* |
| **Volume** | *fixed volume* | *fixed volume* | *fills its container* |
| **Can it be compressed?** | *no* | *barely* | *yes, easily* |

**Check:** read down the gas column. Every row says the particles are free: far apart, moving fast, no shape or volume of their own, easy to compress. Read down the solid column and every row says the opposite.

## 3 · Density and the states same mass, different volume

**Key concept — why a gas has such a low density**

Density is the mass in each unit of volume: $\rho = \dfrac{m}{V}$, which you **recall**. Rearranged, $V = \dfrac{m}{\rho}$.

Take 1 kg of a substance. It has the same particles, so the **same mass**, in every state. What changes is the **volume** they take up:

- **Solid and liquid:** the particles are touching in both, so the same mass takes about the same volume and the two densities are **close**.
- **Gas:** the particles are far apart, with empty space between them, so the same mass takes **hundreds to thousands of times more volume**. The density is tiny.

| Water as… | ice (solid) | liquid water | steam (gas, 100 °C) |
|---|---|---|---|
| density in kg/m$^3$ | 920 | 1000 | 0.60 |

Ice is one of the few solids slightly less dense than its liquid, which is why it floats; for most substances the solid is a little denser.

**Worked example — copy this shape**

**What volume does 1.0 kg of water take up as a liquid, and as steam?**

1. **Write the equation and rearrange it for volume.** $\rho = \dfrac{m}{V}$ so $V = \dfrac{m}{\rho}$
2. **Liquid water:** $V = \dfrac{1.0}{1000} = 0.0010$ m$^3$ (that is one litre, a large bottle)
3. **Steam:** $V = \dfrac{1.0}{0.60} = 1.7$ m$^3$ (a box more than a metre on each side)
4. **Compare:** $1.7 \div 0.0010 = 1700$. The same 1 kg takes about **1700 times the volume** as a gas, because its particles are far apart instead of touching.

**3)** **Calculate** the volume of **2.0 kg** of water (a) as a liquid and (b) as steam. Use the densities in the table. Show the equation, the substitution and the answer with its unit.

$V = \frac{m}{\rho}$. *(a)* $V = \frac{2.0}{1000} = 0.0020\ \text{m}^3$. *(b)* $V = \frac{2.0}{0.60} = 3.3\ \text{m}^3$ *(3.33…).*

**4)** **Calculate** the mass of water in a full bath, which holds $0.15\ \text{m}^3$. Then calculate the volume that mass of water would fill as steam, and compare it with the size of a bathroom, about $30\ \text{m}^3$.

$m = \rho V = 1000 \times 0.15 = 150\,kg$. *As steam,* $V = \frac{m}{\rho} = \frac{150}{0.60} = 250\ \text{m}^3$*, which is more than eight bathrooms. The same particles, the same mass, far more volume.*

> **Check:** for the same mass, the gas volume comes out **thousands of times** the liquid volume.

**5)** **Explain** in terms of the particles, why steam has a much lower density than liquid water.

*In steam the particles are far apart with empty space between them; in liquid water they are touching. So the same mass of water (the same number of particles) takes up a far larger volume as steam, and* $\rho = m/V$ *is far smaller.*

## 4 · Apply it — a new situation, the same model

**6)** **Explain** the following. A syringe with its nozzle sealed is filled with **air**: the plunger can be pushed a long way in. The same syringe filled with **water**: the plunger barely moves. Use the particle model for both.

*Air is a gas, so its particles are far apart with empty space between them. Pushing the plunger moves the particles closer together into that space, so the volume falls a long way (the gas is compressed). Water is a liquid, so its particles are already touching with almost no space between them. They cannot be pushed much closer, so the volume hardly changes and the plunger barely moves.* Model answer: 1 mark for particles far apart in a gas, 1 for the space allowing them to be pushed closer, 1 for particles touching in a liquid, 1 for no space to move into, up to 2 for linking each to the volume change.